SAINT MICHAEL COLLEGE OF CARAGA

VEDIC MATHEMATICS-BASED INSTRUCTION: ITS INFLUENCE ON THE PERFORMANCE AND ATTITUDE TOWARDS MATHEMATICS

A Thesis Presented To
The Faculty of
College of Teacher Education

In Partial Requirements for the Degree
BACHELOR OF SECONDARY EDUCATION MAJOR IN MATHEMATICS

By

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Objectives of the study (qualitative)

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This thesis entitled **"VEDIC MATHEMATICS-BASED INSTRUCTION: IT'S INFLUENCE ON THE PERFORMANCE"** presented and submitted by **Clint Ian U. Gesta, Kaye L. Peralta** in partial fulfillment of the requirements for the degree of **Bachelor of Secondary Education Major in Mathematics** is hereby accepted and recommended for Oral Examination.

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Acknowledgment of Advisors and Professors
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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

Mathematics plays a vital role in science, technology, and innovation. It supports both local education and global competitiveness in math literacy, a key 21st-century competency (OECD, 2023; Situpo et al., 2024). However, results from international assessments such as PISA (2022), as published by the OECD (2023), and TIMSS (2019) reveal that many students continue to struggle with foundational mathematical concepts, underscoring the urgent need for improved instructional methods and deeper cognitive engagement (OECD, 2023).

To address persistent challenges in mathematics education, educators have explored alternatives such as Vedic Mathematics (VM). International studies have demonstrated its effectiveness through distinct outcomes. Kumar (2025) demonstrated that VM techniques significantly enhance calculation speed, particularly in time-constrained environments, by streamlining multi-step arithmetic into rapid mental operations. Parajuli (2020) focused on the structure of basic operations, showing that VM enables one-line mental solutions for addition, subtraction, multiplication, and division, which can be verified quickly and improve learner confidence. Raj Joshi et al. (2025) examined VM's application in teaching multiplication, finding that its pattern-based strategies improved students' symbolic reasoning and mental agility, especially among early secondary learners. These studies suggest that VM supports both procedural fluency and flexible thinking, making it a promising complement to structured instruction and discovery-based learning within the MATATAG Curriculum.

Innovations in teaching and curriculum design can help bridge these gaps. Research confirms that improving math learning outcomes depends on student attitudes, self-efficacy, and motivation (Amerstorfer et al., 2021; Goplani & Jha, 2024; Siller & Ahmad, 2024). Positive

affective-motivational factors are strongly connected to better achievement. On the other hand, a lack of confidence and math anxiety can hinder performance; however, structured approaches such as Direct Instruction (DI) in basic math lead to significant learning gains (Sanchez & Ubalde, 2024; Zetriuslita et al., 2020). Discovery Learning, a component of constructivist education, can foster understanding and critical reasoning when appropriately scaffolded (Alanazi et al., 2024; Çakar Özkan, 2022; De Jong et al., 2023). This instructional synergy aligns with the goals of the MATATAG Curriculum and supports the integration of Vedic Mathematics (VM) to enhance both cognitive and affective outcomes.

In the Philippines, student continue to exhibit low performance in mathematics. Based on reports of different national and international studies indicate that Filipino students lag behind their peers in numeracy and problem-solving skills (Berrame, 2024; Cabuquin & Abocejo, 2023). The lack of resources and negative attitudes in mathematics as persistent constraints to achievement have been identified as cognitive barriers (Mabena et al., 2021; Valderama, 2022). Additionally, local research suggests that factors such as self-efficacy, persistence, and emotional regulation significantly impact student success (Estipona & Delos Santos, 2025; Lazarra, 2025; Peteros et al., 2020).

The Department of Education initiated the MATATAG Curriculum, which advocates inclusive teaching in deconstructing content, fortifying basic abilities, and encouraging contextualized learning. In alignment with these goals, innovative strategies such as Vedic Mathematics (VM) have gained attention for their potential to enhance mathematical engagement and performance. At the junior high school level, Bandala (2023) and Daliva & Deita (2025) found that Filipino junior high school students exposed to Vedic methods enhanced their problem-solving, motivation, and mathematical interest. Complementary local studies support

these findings: Day-Ongao & Tan (2022) reported improved performance and attitudes among students using VM; Taracina (2023) highlighted its inclusivity and effectiveness among Grade 2 SPED learners; and Alabat & Salazar (2025) found VM to be practical, accessible, and motivating from both teacher and student perspectives. However, at the Grade 7 level, empirical evidence is still scarce.

This study investigates the influence of Vedic Mathematics-based instruction, integrated with Direct Instruction (DI) and Discovery Learning (DL), on Grade 7 students' performance and attitudes toward mathematics. By aligning VM strategies with the MATATAG Curriculum, this research aims to provide evidence for effective and culturally responsive instructional practices that foster both mathematical proficiency and positive learning dispositions.

The significance of this study lies in its contribution to the theory and practice of mathematics education. By presenting empirical evidence that can guide teachers regarding adapting culturally and cognitive efficiency strategies for enhancing student learning, this study examines the pedagogical value of Vedic Mathematics in the Philippine education system. Moreover, the results can inform educators in designing interventions and guide curriculum developers in aligning pedagogy with MATATAG goals. Finally, this study seeks to further support mathematics education and learner success in the Philippines. This study, as an instructional paradigm that fuses tradition, efficiency, and inclusivity, should be practiced.

1.2 Review of Literature

Vedic Pedagogical Innovations

Vedic Mathematics (VM) is both a cultural tradition and a practical method for improving mathematical fluency. It is built on sixteen sutras, or guiding principles, along with their sub-formulae, which offer simplified techniques for solving arithmetic and algebra problems. These

strategies encourage mental calculation and pattern recognition. Studies by Raikhola et al. (2020) and Mahajan (2023) show that VM helps students strengthen mental computation, identify patterns, and gain confidence in solving problems.

International research supports the impact of VM on both student performance and mathematical thinking. Ramteke & Vaishnav Rajshree (2019) found that students taught with VM solved problems more efficiently and scored higher than those taught through conventional methods. Saxena (2024) conducted a comparative study of Vedic and conventional methods for arithmetic operations, revealing that VM consistently produced faster calculations and higher accuracy across tasks such as addition, subtraction, multiplication, and division.

Similar results were reported by Parajuli (2020) and Raj Joshi et al. (2025), who observed improved accuracy and better retention of mathematical concepts among learners trained using VM. Subha (2024) helped high school students master multiplication using Vedic algorithms, while Sheron et al. (2022) noted significant test improvements among eighth graders who used VM alongside guided modeling.

These studies consistently demonstrate that VM supports schema automation by transforming abstract computations into logical, visual steps, thereby reducing cognitive load.

Beyond performance, international studies have shown that Vedic Mathematics (VM) also influences students' motivation and attitudes toward the subject. Charde (2024) also noted that students who solved complex tasks through VM became more efficient and had a greater desire to learn. Yogeshwari & Indu (2024) established that VM decreased anxiety and positively changed attitudes, particularly among the more stressed students. Pandey & Sharma (2025) found that visualization and guided practice, used by VM, facilitate the reduction of math anxiety and the development of confidence among learners.

Jangamkote Kiran & Bharadwaj (2024) explored the modern application of Vedic wisdom in education and found that its integration with mindfulness practices fosters emotional regulation and cognitive clarity. Their findings suggest that VM's rhythmic and meditative qualities may contribute to reduced anxiety and improved focus in learning environments.

VM becomes even more effective when combined with structured teaching models. A study using Vedic Mathematics-mediated pedagogy demonstrated that structured, step-by-step instruction can enhance students' problem-solving competency and procedural fluency, particularly when supported by guided practice and conceptual clarity (Sharma et al., 2024). Its procedural strengths align well with Direct Instruction (DI), a teacher-led approach that emphasizes explicit modeling and guided practice. Sahu et al. (2024) found that VM paired with DI was more effective in solving semi-routine problems than the Concrete-Representation-Abstract (CRA) model. VM also complements Discovery Learning (DL), which encourages students to explore and build knowledge through inquiry.

While empirical studies on VM within DL frameworks remain limited, its alignment with Constructivist Learning Theory suggests potential for inquiry-based integration (Pimpalkar, 2025). These findings suggest that VM is a flexible tool that can be integrated into both DI and DL frameworks to support cognitive and emotional development.

These findings collectively indicate that VM can be integrated into both Direct Instruction and Discovery Learning, making it a versatile pedagogical tool that supports both procedural fluency and inquiry-based learning.

Vedic Mathematics (VM) has also been explored in the fields of computing and technology, showing that its value extends beyond classroom instruction. S. et al. (2022) and Shetkar & Koli (2023) found that VM sutras can improve digital multiplier architecture, suggesting

practical applications in hardware design. Although its direct impact on learning outcomes is harder to measure, VM's mathematical reliability makes it suitable for STEM programs. Kumra & Garg (2023) noted that VM's pattern-based reasoning facilitates numeracy in contemporary educational contexts. These findings collectively indicate that VM's sutras extend to computing and STEM applications, showing its adaptability beyond pedagogy

In the Philippines, research on Vedic Mathematics (VM) is still limited but steadily growing. Bandala (2023) found that students who received instructions on Vedic Mathematics Techniques (VMT) were more confident and focused when compared to students who underwent traditional methods. Daliva & Deita (2025) noted improvements in computing and classroom performance of Grade 7 students who were taught VM. Similarly, Roldan (2025) demonstrated that Vedic Math techniques significantly improved the numerical skills of Filipino learners, reinforcing VM's role in enhancing computational fluency in local classrooms. Building on these findings, other local researchers have started to explore how VM can enhance motivation and inclusivity in Philippine classrooms.

Alova (2020a) added that VM promotes mental economy by reducing unnecessary cognitive effort through efficient strategies. In a related study, Alova (2020b) demonstrated how Vedic algorithms connect procedural fluency with conceptual understanding, drawing parallels to algebraic concepts such as Pascal's theorem. These findings collectively align with cognitive load and constructivist learning theories, emphasizing that learners internalize mathematical schemas through structured, pattern-based reasoning rather than rote memorization.

Further studies in the Philippines add depth to the evidence on Vedic Mathematics (VM). According to Day-Ongao & Tan (2022), VM enhanced problem-solving and positively influenced the attitude toward math. They said that VM converts abstract calculations to intuitive patterns

based on the natural rhythm of the sutras. At the primary level, Taracina (2023) discovered that VM enhanced simple operations in Grade 2 SPED learners, indicating its role in inclusive education.

Alabat & Salazar (2025) examined the perception of teachers and students towards the Cauchy-Vedic Mathematics in the classroom. They discovered that VM was perceived as useful, rational, and helpful. However, they emphasized that practical teacher training is necessary to achieve conceptual sophistication and computing efficiency (Alabat & Salazar, 2025). All in all, studies confirm the availability and applicability of VM to various grades, such as SPED and junior high school (Day-Ongao & Tan, 2022; Taracina, 2023).

Together, these Philippine studies reinforce VM's accessibility, showing its promise across learning levels—from SPED to junior high—while underlining the need for structured teacher preparation for consistent implementation. These findings collectively indicate that local studies confirm VM's effectiveness in improving computational fluency, inclusivity, and motivation, though teacher training and curriculum integration remain challenges.

Nationally, DepEd Memorandum No. 54 (2025), formally recognized VM's educational value by endorsing the MATH-Inic Vedic Mathematics National Challenge, which signifies VM as an educational potential. Similarly, academic activities like the 6th Philippine Vedic Mathematics Olympiad (PNVMO) (2024) and LSPU's CTE-Sipnayan (2024) support the use of the VM as a system growing in adaptation as a learning innovation and supplemental teaching. However, most of the efforts remain limited as extracurricular activities are more focused on enrichment rather than integration as a part of the core of the curriculum.

A review of both global and local studies highlights five recurring themes about Vedic Mathematics (VM). First, VM strengthens procedural skills essential for higher-level math (Bandala, 2023; Parajuli, 2020; Raj Joshi et al., 2025; Sahu et al., 2024; Sharma et al., 2024).

Second, it makes math more approachable, boosting motivation and self-confidence (Charde, 2024; Daliva & Deita, 2025; Pandey & Sharma, 2025; Yogeshwari & Indu, 2024). Third, without proper guidance, VM risks being reduced to shortcuts, limiting deeper understanding (Alabat & Salazar, 2025; Mahajan, 2023; Ramteke & Vaishnav Rajshree, 2019). Fourth, its success depends on teacher readiness, student preparation, and curriculum alignment (Alabat & Salazar, 2025; Pimpalkar, 2025). Fifth, most studies are small and short-term, making it hard to generalize findings (Mahajan, 2023; Ramteke & Vaishnav Rajshree, 2019; Taracina, 2023).

Taken together, these insights show that VM supports both cognitive and emotional learning. When paired with structured teaching, it helps students learn more effectively and fosters a greater sense of confidence. However, without clear guidance, it risks being oversimplified. Despite promising results, research gaps remain. Few studies have explored how VM fits into the MATATAG Curriculum or its impact on advanced topics, such as algebra, geometry, and data literacy. Further research is needed on the interactions between teacher practices, student attitudes, and curriculum reforms in Philippine secondary schools.

In conclusion, VM is a versatile approach that improves math performance and student attitudes. While international studies offer strong support, Philippine research is still in its early stages of development. Continued investigation is needed to integrate VM into national reforms. By blending traditional wisdom with modern innovation, VM demonstrates promise as a culturally relevant and effective model for enhancing math education within the MATATAG framework.

Instructional Approaches in Mathematics Education

The pedagogy of mathematics determines how the subject is understood, comprehended, and valued by students. There are two well-known instructional models: Direct Instruction (DI) and the Discovery Learning (DL) model. DI is an instructional model that focuses on teachers and

gives great importance to clear explanations, guided practice, and skill mastery (Mason & Otero, 2021). DL, in turn, promotes the process of knowledge building through discovery and investigation (Widana & Handayani, 2022).

International studies confirm that DI enhances mathematics performance by improving procedural fluency and retention through structured explanations and repetition (Mason & Otero, 2021; Shammas, 2023; Stockard et al., 2020; Yaghmour & Obaidat, 2022).

Winarsih et al. (2019) found that DI and problem-based learning both improved performance among millennial learners. Hammond (2022) explored attitudes toward DI in Australian schools, showing positive perceptions, especially in primary settings. Nur et al. (2023) demonstrated that DI with task-sheet-based learning increased motivation during the COVID-19 crisis.

Eratay (2020) showed that DI was effective in teaching leisure skills to young individuals with intellectual disabilities. Zetriuslita et al. (2020) found a beneficial effect of DI on self-regulation and autonomy when combined with digital tools, such as GeoGebra.

While the studies of De Jong et al. (2023) and Alanazi et al. (2024) have discovered that DL encourages the process of conceptual and creative thinking, Umamah et al. (2024) demonstrated that DL assisted by Focusky media improved historical analysis and learning outcomes, showing DL's adaptability across disciplines.

Muhammad & Juandi (2023) conducted a bibliometric review of DL in mathematics from 1968–2023, confirming its sustained relevance and evolution in math education. Tshering et al. (2024) found that DL increases persistence, and Widana & Handayani (2022) found that it increases intrinsic motivation and teamwork.

Suryanti & Budijastuti (2020) found that guided discovery problem-posing enhanced science process skills in elementary learners. Le & Nguyen (2024) noted that discovery enhances curiosity. These findings collectively indicate that DL fosters creativity, persistence, and intrinsic motivation, making it adaptable across disciplines.

Although they are different, the two approaches are complementary to one another: DI develops the primary fluency, and DL supports more profound knowledge De Jong et al. (2023). They would be able to reinforce numeracy and problem-solving skills when combined with Vedic Mathematics (VM), as proposed in the MATATAG Curriculum.

Rahmayanti & Kwalat (2024) and Lu et al. (2025) have also noted that the use of DI and DL in combination is more accurate and insightful, and guided modeling with the use of DL enhances cognitive development. Siregar et al. (2024) have demonstrated that DI and DL, when integrated with computer assistance, can have a significant impact on improving mathematics performance. Sun et al. (2022) demonstrated that DI, coupled with collaborative tasks of DL, results in greater accuracy and involvement. These findings collectively indicate that DI and DL complement each other, balancing fluency and conceptual depth.

Finland and Singapore are examples of countries worldwide that have implemented hybrid features, combining structured instruction with inquiry-based work. The CPA approach (Concrete-Pictorial-Abstract) of Singapore, as reviewed by Main (2024) is an example of such a balance, in which students transition to abstract reasoning while maintaining conceptual rigor and fluency to solve problems. Mae et al. (2025) also noted that both Finland and Singapore emphasize conceptual knowledge based on inquiry. However, they have a structured progression as a manifestation of a global move towards instructional hybridity.

Areepattamannil et al. (2020) conducted a cross-national study across 66 countries showing that teacher-directed and inquiry-based instruction both contribute to positive science dispositions. These findings collectively indicate that global practices integrate DI and DL to balance structure and inquiry, reflecting a trend toward instructional hybridity.

These results provide evidence that Direct Instruction (DI) and Discovery Learning (DL) contribute to the control of cognitive load in different ways. DI facilitates working memory, which entails having a structured approach, which minimizes extraneous cognitive load and supports schema acquisition (Stockard et al., 2020; Sweller et al., 2019). While DL promotes creativity and the transfer of knowledge by optimizing germane load and encouraging active engagement (De Jong et al., 2023; Ouwehand et al., 2025). They combine to create a balanced method of instruction that aligns with the purposes of the MATATAG Curriculum (Kim et al., 2025).

Affective outcomes further support the combined use of DI and DL. Hidayah & Prayoga (2021) found that DL reduces math anxiety by shifting the focus from valuation to exploration. Zetriuslita et al. (2020) concluded that structured feedback in DI lowers frustration. Murillo & Denis (2019) and Kevin et al. (2021) reported that DL boosts motivation and collaboration, while DI provides confidence. Le & Nguyen (2024) noted that discovery enhances curiosity, and Rahmayanti & Kwalat (2024) confirmed that DI's explicit sequencing improves learner satisfaction. These findings collectively indicate that DI and DL balance cognitive load differently and together improve both confidence and motivation.

Philippine research reflects similar trends to international studies, affirming the effectiveness and adaptability of Direct Instruction (DI) and Discovery Learning (DL) across diverse educational settings. Direct Instruction (DI) has been shown to enhance fluency and comprehension in problem-solving tasks, and its structured nature has been leveraged to address

(Bandahala, 2024a), resource allocation issues in public schools (Aguhayon et al., 2023). Studies also confirm its foundational impact in early education, with explicit instruction improving reading comprehension among Grade 1 pupils (Lucero & Vargas, 2021) and foundational skills in kindergarten learners (Cadungog & Rollo, 2024). DI-based formative assessments have led to improved accuracy and satisfaction (Dela Peña & Doronio, 2025), while explicit instruction has increased engagement in mathematics (Sanchez & Ubalde, 2024). According to the study of Magsalay (2019) found that DI outperformed other models in promoting achievement, particularly in problem-solving.

Discovery Learning (DL), on the other hand, has demonstrated strong motivational and cognitive benefits. DL-based modules have been shown to increase motivation and comprehension in General Mathematics Mae et al. (2023). DL has also been applied effectively to trigonometric identities (Bernido et al., 2019), and its strategies have improved student engagement (Murillo & Denis, 2019). DL's inclusive potential is evident in its success with Grade 7 struggling learners (Cellona & Padrillan, 2025), and its role in enhancing higher-order thinking and accuracy has been confirmed (Golingan & Sulatra, 2022). These findings collectively indicate that Philippine studies confirm DI's role in fluency and DL's role in motivation, both contributing to improved performance.

Several studies emphasize the complementary nature of DI and DL. Jude & Joaquin (2022) demonstrated that guided discovery paired with explicit instruction improved problem-solving, while Clores et al. (2025) highlighted their interrelation. Dela Peña & Doronio (2025) found that DI combined with formative assessment led to deeper learner satisfaction, and Medina & Baraquia (2023) confirmed that teacher-led inquiry produced more meaningful learning experiences. Bibon (2022) concluded that instructional practices—whether DI or DL—significantly influence academic

achievement. These findings suggest that instructional effectiveness depends on how teachers blend both methods in response to student readiness, task complexity, and curricular goals.

In practice, DI and DL are most effective when used together. Santyasa et al. (2019) found that combining them enhances critical thinking and collaboration, while Lethulur et al. (2025) showed that structured discovery yields better retention than either method alone. Philippine evidence supports this integration: Niman et al. (2024) reported that combining DI and DL improves understanding, motivation, and retention, and Flores et al. (2025) found that discovery supported by explicit instruction strengthens problem-solving and social awareness among daycare pupils. Sturrock et al. (2025) observed that both DI and inquiry-based instruction are pragmatically integrated in authentic classroom settings, although implementation varies across primary and secondary levels. These findings collectively indicate that DI and DL, when blended, enhance problem-solving, retention, and learner satisfaction in Philippine classrooms.

Despite these promising findings, research gaps remain. Most Philippine studies focus on short-term outcomes and overlook specific competencies such as algebraic reasoning and problem-solving (Famulagan, 2024; Magayon & Tan, 2020). Few studies have examined the long-term effects of integrating DI and DL on teacher practices and student success. There is also limited research on combining these approaches with Vedic Mathematics (VM) under the MATATAG Curriculum (Roldan, 2025). Comparative studies on the transition from arithmetic to algebra in Grade 7 are notably lacking, even though this stage is critical for both cognitive and emotional development (Legal P., 2024).

Addressing these gaps will provide stronger evidence on how teachers can effectively integrate DI and DL with VM to improve student performance, reduce anxiety, and foster lasting mathematical confidence (Hidayah & Prayoga, 2021; Sweller et al., 2019; Zetriuslita et al., 2020).

Overall, the literature shows that DI and DL are equally important for effective mathematics teaching. DI provides precision and structure, while DL fosters creativity and conceptual insight (De Jong et al., 2023; Sweller et al., 2019). When integrated with Vedic Mathematics (VM), this "structured flexibility" reflects the vision of the MATATAG Curriculum (Roldan, 2025), where learning is both guided and self-directed. By merging the discipline of DI with the dynamism of DL, teachers can create classrooms that nurture accuracy, reflection, and innovation—skills essential for 21st- century mathematics education (Kim et al., 2025; Situpo et al., 2024).

Learners' Attitude toward Mathematics

How students feel about mathematics plays a significant role in their academic success. Their attitude influences how confident, motivated, and engaged they are in learning (Gjicali & Lipnevich, 2021; Schunk & DiBenedetto, 2021). When students enjoy math and believe in their abilities, they participate more actively and understand the material better. However, when they feel anxious or discouraged, they often avoid the subject and struggle to keep up (Li et al., 2023; Sturm & Bohndick, 2021).

Global research shows a strong link between math attitudes and student outcomes. Li et al. (2023) found that two key indicators of success are self-confidence and interest in mathematics. According to Schunk & DiBenedetto (2021), self-efficacy enables students to persevere in the face of complex issues. Hwang & Son (2021) and Wen & Dubé (2022) demonstrated that enjoyment and engagement are increased with positive support. Lehocká & Kmeťová (2025) emphasized that student motivation and attitude are critical determinants of mathematics achievement, particularly during early adolescence. Aquilina et al. (2025) proposed a model in which enjoyment, perceived value, and usefulness condition the student's effort. According to Alanazi et al. (2024) and Tshering et al. (2024), this model has been found to improve

motivation and attitudes in discovery-based learning. Rahmayanti & Kwalat (2024) ensured that it also enhances self-efficacy and commitment. These findings collectively indicate that global studies confirm the strong influence of confidence, enjoyment, and motivation on mathematics achievement.

Vedic Mathematics (VM) has shown consistent benefits in shaping learners' emotional engagement and confidence in mathematics. International studies highlight VM's ability to reduce anxiety and foster motivation through rhythmic, pattern-based reasoning and mental economy (Jangamkote Kiran & Bharadwaj, 2024; Pandey & Sharma, 2025; Yogeshwari & Indu, 2024). These techniques convert abstract calculations into intuitive steps, supporting emotional regulation and cognitive clarity (Mahajan, 2023).

Pedagogical approaches show that collaboration and exploration have nurtured significant positive attitudes towards mathematics. Direct Instruction (DI) and Discovery Learning (DL) also play pivotal roles in shaping learners' attitudes toward mathematics. Globally, DI's structured feedback and explicit sequencing have been shown to reduce frustration and improve learner satisfaction (Rahmayanti & Kwalat, 2024; Zetriuslita et al., 2020), while DL fosters curiosity, intrinsic motivation, and collaborative engagement (Le & Nguyen, 2024; Murillo & Denis, 2019; Widana & Handayani, 2022). Amerstorfer et al. (2021) found that discovery learning boosts intrinsic motivation and emotional engagement, especially in cooperative settings. Siller & Ahmad (2024) justified this opinion that DL boosts the motivation of students.

Goplani & Jha (2024) demonstrated that anxiety is reduced and confidence is built when it is possible to decrease the number of tedious calculations and develop pattern-based reasoning. These international perspectives highlight that mathematics attitudes are not fixed traits but teachable dispositions—molded through supportive instruction, interactive design, and positive

classroom culture. These findings collectively indicate that VM, DI, and DL all contribute to reducing anxiety and building confidence, shaping mathematics attitudes as teachable dispositions.

Within the Philippine context, student motivation and performance are closely tied to their attitudes toward math. Caputol & Chan (2024) found that mathematics attitude mediates the relationship between motivated learning strategies and anxiety, highlighting the emotional complexity behind student engagement. Similarly, Gamit (2025) explored how parental and teacher support correlate with math attitudes and performance among Grade VI pupils, though the study was descriptive in nature. Camacho (2025)) and Malinao (2025) noted that confident and motivated students perform better in math compared to their peers who are intimidated by the subject. According to Tamayo (2021), persistence was influenced by confidence and engagement. In contrast, outcomes were associated with affective disposition and study habits (Capuno et al., 2019). As demonstrated by Carcueva (2024), emotional balance plays a crucial role in the success of learning.

In the Philippine setting, DI has increased engagement and foundational confidence (Lucero & Vargas, 2021; Sanchez & Ubalde, 2024), while DL has supported persistence and emotional balance, especially among struggling learners (Camacho, 2025; Cellona & Padrillan, 2025). Building on these findings, Clores et al. (2025) discovered that cross-disciplinary learning of literacy and math lessons enhances confidence and persistence, implying that emotional growth is facilitated by cross-disciplinary instruction. In their study, Medina & Baraquia (2023) have found that math lessons built through inquiry skills improved attitudes, resulting in increased confidence and persistence.

Aguhayon et al. (2023) stated that an increase in motivation and a decrease in anxiety follow the application of contextualized and scaffolded teaching in relation to the MATATAG Curriculum. Laranang & Bondoc (2020) discovered that self-efficacy also contributes to the prevention of math anxiety. These studies reinforce the MATATAG Curriculum's emphasis on combining structure and exploration to build confident, motivated learners who view mathematics as both accessible and meaningful.

VM has improved computing performance and motivation among Grade 7 students (Bandala, 2023; Daliva & Deita, 2025), and has been perceived as useful and rational by both teachers and learners (Alabat & Salazar, 2025). Reyes (2025) demonstrated that educational games, such as DAMATH, enhance enjoyment and engagement. Meanwhile, Aujero & Fuentebilla (2025) found that a structured teaching approach significantly reduced math anxiety and improved critical thinking skills among Grade 12 STEM students.

Day-Ongao & Tan (2022) reported that students using Vedic Mathematics not only performed better but also developed more positive attitudes toward the subject. They described VM as rhythmic and pattern-based, making it a more motivating approach. Meanwhile, Taracina (2023) demonstrated its inclusive potential among SPED learners. Collectively, these findings suggest that VM supports both procedural fluency and affective growth, making it a culturally responsive tool for inclusive mathematics instruction.

Overall, Philippine studies suggest that instructional design, teacher support, and classroom environment have a strong influence on students' emotions and motivation (Aguhayon et al., 2023; Camacho, 2025; Medina & Baraquia, 2023). These factors have been shown to enhance enjoyment, engagement, and persistence (Clores et al., 2025; Day-Ongao & Tan, 2022; Reyes, 2025).

Much of the Philippine research on mathematics attitudes remains descriptive or correlational in nature, with limited longitudinal or intervention-based designs. For example, Caputol & Chan (2024) and Gamit (2025) employed correlational methods that highlight short-term associations but do not capture long-term attitudinal change. As a result, while short-term affective outcomes are well-documented, sustained changes in mathematical attitudes over time are rarely measured. This gap underscores the need for more robust empirical studies that examine how instructional reforms—such as those under the MATATAG Curriculum—impact learners' emotional development and long-term confidence in mathematics.

These findings collectively indicate that Philippine studies confirm the influence of instructional design, teacher support, and VM integration on student attitudes, though most remain short-term and descriptive.

Both global and national literature have positive perceptions of mathematics and associate motivation, persistence, and achievement in mathematics (Camacho, 2025; Li et al., 2023; Malinao, 2025; Schunk & DiBenedetto, 2021). The correlation between emotional disposition and academic achievement shows the importance of teaching strategies that encourage pleasure and self-efficacy and confidence of learners, not only computational skills (Gjicali & Lipnevich, 2021; Hidayah & Prayoga, 2021; Tamayo, 2021). Nevertheless, there are no Philippine studies that have directly evaluated the combined effect of Vedic Mathematics, Direct Instruction (DI), and Discovery Learning (DL) on student attitudes within a high school setting. While Alanazi et al. (2024), Clores et al. (2025), and Rahmayanti & Kwalat (2024) highlight the influence of instructional strategies on affective outcomes, they do not isolate or longitudinally assess these specific methods in secondary education.

This study examines the effect of integrating Vedic Mathematics with Direct Instruction (DI) and Discovery Learning (DL) on the attitudes of Grade 7 students toward mathematics. By focusing on both emotional and cognitive growth, it supports the MATATAG Curriculum's mission to build confident, motivated learners who see math as helpful and engaging. The findings aim to strengthen affective learning theory and support national education reforms by enhancing basic math skills and empowering students to achieve success.

Determinants of Mathematics Performance

One should be aware of the factors that influence the performance of students studying mathematics to improve learning outcomes and teaching practices. These are the known determinants, which include the quality of instructions, the learner's disposition, the emotional component, and system support.

International studies consistently highlight that mathematics achievement is shaped by a combination of instructional quality, learner disposition, emotional regulation, and systemic support. Lavelle-Hill et al. (2024) emphasized that instructional techniques and teacher clarity are critical for long-term performance. Ulkhaq (2023), referencing PISA data, found that teacher feedback and classroom climate are stronger predictors of mathematics success than socioeconomic background. Rose et al. (2022) identified instructional interruptions and disparities during the pandemic as key contributors to global inequality in achievement.

Learner disposition and emotional preparedness also play a central role. Hwang & Son (2021) revealed that students with positive attitudes and self-perception tend to perform better. Skagerlund et al. (2025) demonstrated that anxiety and regulation strategies significantly affect outcomes. Abín et al. (2020) and Schunk & DiBenedetto (2021) linked self-efficacy and motivation to persistence and improved learning. Similarly, Wakhata et al. (2024) and Sturm & Bohndick

(2021) showed that students' emotional responses to problem-solving tasks are closely tied to their success.

Systemic and environmental factors further influence performance. Li et al. (2025) emphasized the importance of teacher-student relationships and classroom climate in fostering motivation and achievement. The OECD (2023) underscored mathematical literacy as a foundational skill for national competitiveness and informed citizenship. Djekourmane et al. (2025) demonstrated that parental involvement enhances mathematics self-efficacy, which in turn improves achievement among Southeast Asian learners.

Recent studies also emphasize the interaction between cognitive and affective domains. Li et al. (2023) identified motivation and engagement as latent predictors of mathematics achievement across countries. Lehocká & Kmeťová (2025) stressed that student attitude is a critical determinant, particularly during early adolescence. Hwang & Son (2021) and Wen & Dubé (2022) found that a positive attitude toward mathematics enhances both conceptual understanding and performance among Asian students.

Although instructional strategies such as Vedic Mathematics (VM), Direct Instruction (DI), and Discovery Learning (DL) have been shown to influence affective outcomes, most global studies treat cognitive and emotional factors separately and rarely examine their interaction within specific teaching methods. This gap suggests the need for more integrative approaches that address both domains simultaneously.

These findings collectively indicate that international studies highlight instructional quality, emotional preparedness, and systemic support as key determinants of mathematics performance, but often fail to integrate cognitive and affective domains.

In the Philippines, similar determinants affect mathematics performance, though challenges in teaching capacity and resource availability often exacerbate their impact. The country has consistently ranked among the lowest performers in international assessments, with PISA 2022 results showing that Filipino 15-year-olds scored an average of 355 points in mathematics, compared to the OECD average of 472. Only 16% of students reached Level 2 proficiency—the baseline for functional mathematical literacy—and almost none attained top performance levels (OECD, 2023).

Bernardo et al. (2022) attributed these outcomes to inequitable access and weak implementation of reform policies. Luzano (2024) and Acido et al. (2023) found that socioeconomic status remains a strong predictor of achievement, while Berrame (2024) noted that classroom realities often limit the success of curriculum changes. In response, local studies have explored contextualized and technology-based interventions. Mendiola & Estonanto (2022) reported that localized materials improve comprehension and retention. Flores et al. (2024) found that instructional flexibility correlates with better student outcomes. Etcuban & Pagaran (2025) demonstrated that blended learning enhances mastery, and Borabo & Dio (2025) showed that remediation using SIM improves calculation accuracy.

Emotional and psychological factors have also been examined. Estipona & Delos Santos (2025) found that well-being and life satisfaction predict mathematics performance. Cabigas & Cabalquinto (2025) emphasized the role of attitude and parental involvement, while Sulpico et al. (2025) demonstrated that gamified learning boosts motivation and performance in low-resource schools. Dicdiquin et al. (2023) highlighted the long-term impact of teacher professional development on classroom delivery and student outcomes. Collectively, these studies suggest

that teaching quality, student motivation, and parental involvement have a synergistic effect on mathematics performance, though such benefits are often short-term and context-specific.

Philippine literature also affirms the interaction between cognitive and affective factors. confirmed that anxiety regulation, self-efficacy, and emotional resilience contribute to persistence and achievement. However, most studies do not evaluate how these variables interact within specific instructional models.

While VM, DI, and DL are recognized as promising strategies, Philippine research has yet to assess their combined long-term impact. Existing studies tend to isolate these methods or focus on immediate outcomes (Berrame, 2024; Dicdiquin et al., 2023; Sulpico et al., 2025), leaving a gap in understanding how reforms like the MATATAG model can be tailored to diverse learners.

Recent local literature suggests that VM's pattern-based reasoning reduces anxiety and fosters confidence, DI's structured feedback enhances satisfaction, and DL's inquiry-based design boosts motivation and engagement. These affective outcomes are critical, as they directly influence mathematics performance and align with the MATATAG Curriculum's goal of building confident, motivated learners.

Considering these determinants, the current study examines the potential of combining VM, DI, and DL to optimize the application of cognitive and affective spheres in mathematics learning. By integrating these approaches, the study aims to contribute to affective learning theory and support national education reforms that promote both academic mastery and emotional resilience.

Synthesis of Review of Related Literature

The literature suggests that learning mathematics is most effective when instruction fosters both fluency and emotional engagement (Gjicali & Lipnevich, 2021; Schunk & DiBenedetto, 2021). This dual focus supports learners cognitively and affectively, aligning with the MATATAG Curriculum's goal of nurturing confident, self-motivated problem solvers.

Theme 1: Vedic Mathematics as a Pedagogical Innovation

Shows that Vedic Mathematics (VM) improves computational efficiency and motivation. It does this by turning abstract arithmetic into pattern-based reasoning (Goplani & Jha, 2024; Mahajan, 2023; Raikhola et al., 2020). In the Philippines, VM has been shown to improve attention, reduce anxiety, and increase classroom participation among Grade 7 students (Bandala, 2023; Daliva & Deita, 2025; Day-Ongao & Tan, 2022). These findings support the role of VM in inclusive and culturally responsive teaching and learning.

Theme 2: Instructional Approaches in Mathematics Education

Highlights the roles of Direct Instruction (DI) and Discovery Learning (DL). DI offers structural clarity and precision, while DL encourages exploration and conceptual insights. Together, they balance precision and creativity, supporting both procedural fluency and more profound understanding (De Jong et al., 2023; Mason & Otero, 2021; Widana & Handayani, 2022). Philippine studies confirm that DI and DL improve performance, motivation, and inclusivity, especially among struggling learners and in resource-limited schools (Aguhayon et al., 2023; Bandahala, 2024b; Cellona & Padrillan, 2025; Clores et al., 2025).

Theme 3: Learners' Attitudes Toward Mathematics

Explores the impact of emotions on academic performance. Positive emotions such as enjoyment, confidence, and curiosity lead to greater persistence and better achievement outcomes (Hwang & Son, 2021; Li et al., 2023; Wen & Dubé, 2022). In the Philippine context,

studies have shown that scaffolded instruction, contextualized modules, and educational games, such as DAMATH, improve student attitudes and reduce anxiety (Aujero & Fuentebilla, 2025; Camacho, 2025; Medina & Baraquia, 2023; Reyes, 2025).

Theme 4: Determinants of Mathematics Performance

Mathematics performance is shaped by instructional quality, emotional preparedness, and systemic support. International studies have highlighted the roles of teacher clarity, classroom climate, and emotional regulation in predicting achievement (Lavelle-Hill et al., 2024; Skagerlund et al., 2025; Ulkhaq, 2023). Philippine research echoes these findings, noting that differentiated instruction, parental involvement, and gamified learning have a positive influence on performance (Cabigas & Cabalquinto, 2025; Dela Peña & Doronio, 2025; Sulpico et al., 2025). However, most local studies are short-term and do not examine how cognitive and emotional factors interact within specific instructional models, such as VM, DI, and DL.

These themes suggest that instructional design should strike a balance between clarity, creativity, and confidence. Integrating VM into DI and DL creates a supportive structure while fostering conceptual understanding. Evidence suggests these combined approaches enhance numeracy skills, student engagement, and resilience. However, research remains fragmented. There is no substantial evidence base for these practices, despite studies from both international sources and the Philippines supporting their use. Little empirical research has examined how these methods work together in real classrooms or their lasting effects after short-term interventions.

This study builds on this synthesis by exploring how an integrated pedagogy of Vedic Mathematics (as an intervention) and Direct Instruction and Discovery Learning (as experimental groups) affects the academic achievements and attitudes of Grade 7 students. By combining cognitive and emotional aspects, the study aims to support the MATATAG Curriculum's goal of developing confident, self-motivated problem solvers.

1.3 Theoretical Framework

This study is anchored in three interrelated theories: Constructivism (Jean Piaget, 1936; Jerome Bruner, 1960), Cognitive Load Theory (John Sweller, 1988), and the Theory of Attitude Formation and Change (Daniel Katz, 1960; Fishbein & Ajzen, 1975). Each offers a distinct lens for understanding how students acquire knowledge, develop mathematical skills, and form attitudes toward learning. Together, they support the integration of Vedic Mathematics (VM), Direct Instruction (DI), and Discovery Learning (DL) as a combined teaching approach to enhance both performance and mindset in mathematics.

Constructivism (Piaget, 1936; Bruner, 1960) emphasizes that learners actively build knowledge by connecting new experiences to prior understanding. In this study, it supports the use of Discovery Learning in exploring Vedic Mathematics patterns, while also recognizing the role of teacher scaffolding in providing feedback and conceptual support.

Direct instruction, on the other hand, is a teacher- centered approach that is organized, fluent, and accuracy-oriented in contrast to Discovery Learning, which promotes critical thinking and problem-solving. Mathematics contains a collection of systematic concepts, which serve as cognitive anchors and help students learn essential concepts. Constructivism, therefore, bridges DI and DL and creates a balance between structure and exploration.

Cognitive Load Theory (Sweller, 1988) explains that effective instruction reduces unnecessary mental effort and strengthens schema formation. Vedic Mathematics aligns with this principle by simplifying calculations, which complements Direct Instruction and Discovery Learning in easing working memory demands and improving accuracy.

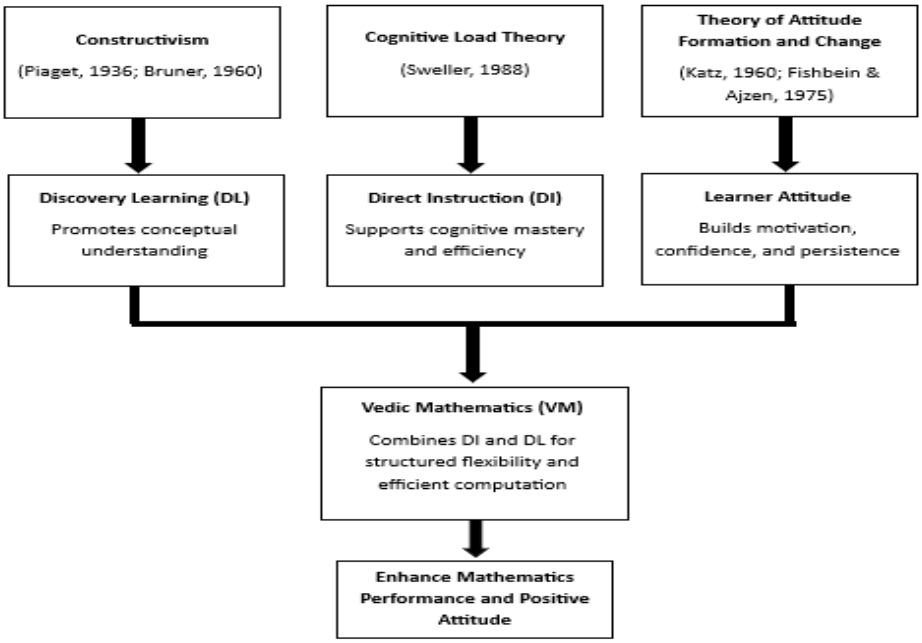
The Theory of Attitude Formation and Change (Katz, 1960; Fishbein & Ajzen, 1975) explains how beliefs, emotions, and experiences shape students' motivation and persistence. In this study, it frames how Vedic Mathematics, supported by Direct Instruction and Discovery Learning, may reduce anxiety, build confidence, and foster positive attitudes toward mathematics.

Together, these three theories provide a comprehensive foundation for the study. Constructivism emphasizes active engagement and exploration, Cognitive Load Theory ensures efficiency in learning through reduced mental effort, and the Theory of Attitude Formation and Change highlights the emotional and motivational dimensions of learning. Their integration justifies the use of Vedic Mathematics through Direct Instruction and Discovery Learning as a means to improve both performance and attitudes in mathematics.

These models will inform the analysis of performance and attitudes in the study, focusing on the integration of Vedic Mathematics, Direct Instruction, and Discovery Learning. Their focus is on ensuring that the outcomes of success encompass both the quality of teaching and the depth of the approach in promoting confidence, engagement, and resilience.

Figure 1. Theoretical Framework of the Study

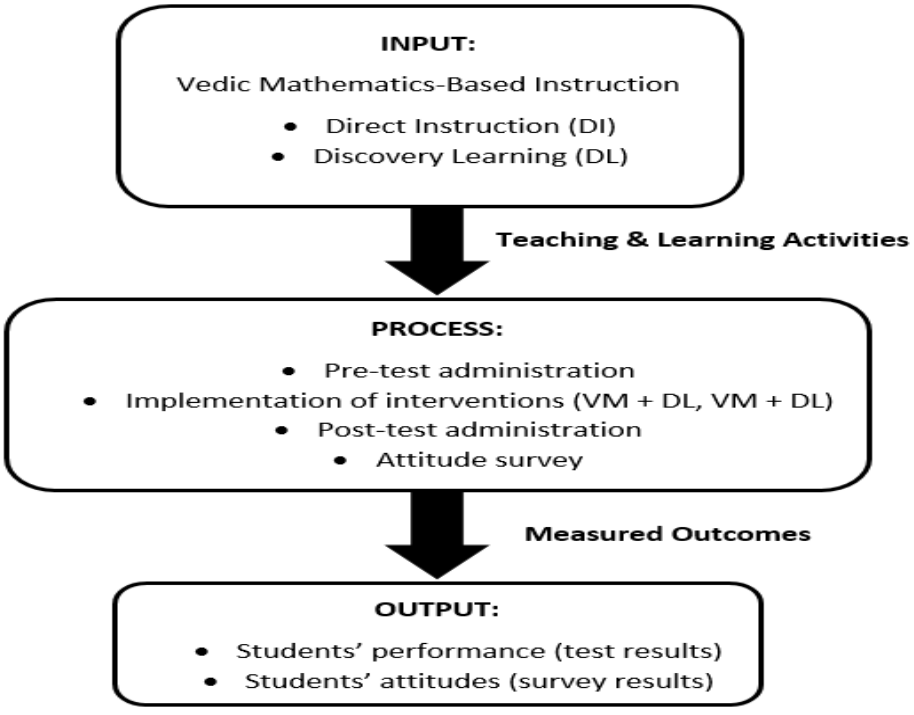
Figure 1 illustrates the interconnection of the three theories to support the study. Constructivism (Piaget, 1936; Bruner, 1960) links to Discovery Learning, emphasizing conceptual understanding through exploration. Cognitive Load Theory (Sweller, 1988) supports Direct Instruction by promoting efficient learning and mastery. The Theory of Attitude Formation and Change (Katz, 1960; Fishbein & Ajzen, 1975) pertains to learner attitude, emphasizing motivation and confidence. Together, these frameworks explain how Vedic Mathematics combines structure and inquiry to strengthen both mathematical performance and a favorable disposition toward learning.



Together, these theories justify integrating Vedic Mathematics with Direct Instruction (DI) and Discovery Learning (DL), providing a balanced foundation that supports both cognitive mastery and positive affective outcomes in mathematics learning.

1.4 Conceptual Framework

Figure 2. The conceptual framework shows the study's structure using the Input–Process–Output (IPO) model. The input includes Vedic Mathematics–based lessons taught through Direct Instruction (DI) and Discovery Learning (DL), along with a control group using traditional teaching. The process covers the administration of pre- tests, the implementation of interventions, and the administration of post-tests and attitude surveys. The Output reflects the measured outcomes—students' performance, their attitudes toward mathematics, and the comparative results of the groups. This framework illustrates the logical flow of the research from inputs to outcomes.



1.5 Statement of the Problem

Students often struggle with operations on polynomials, particularly during the transition from arithmetic to algebra. Traditional teaching methods have not sufficiently addressed these difficulties, leading to low performance and negative attitudes toward mathematics. This study will investigate whether Vedic Mathematics, taught through Direct Instruction and Discovery Learning, can improve Grade 7 students' performance and attitudes in mathematics.

1. What are the pre-test and post-test scores of students in mathematics before and after the application of Vedic Mathematics through Direct Instruction and Discovery Learning?
2. Is there a significant difference in the mathematics performance of students using Vedic Mathematics who are taught through Direct Instruction compared with those taught through Discovery Learning?
3. Is there a significant difference in students' attitudes toward mathematics using Vedic Mathematics when taught through Direct Instruction compared with those taught through Discovery Learning?
4. Is there a significant relationship between students' mathematics performance and their attitudes toward mathematics after being exposed to Vedic Mathematics through Direct Instruction and Discovery Learning?

1.6 Hypothesis

Ho_1: There is no significant difference in mathematics performance between students taught through Direct Instruction and Discovery Learning.

Ho_2: There is no significant difference in attitude toward mathematics between students taught through Direct Instruction and Discovery Learning.

Ho_3: There is no significant relationship between students' mathematics performance and their attitudes toward mathematics after being exposed to Vedic Mathematics through Direct Instruction and Discovery Learning.

1.7 Scope and Limitations

This study focuses on the effect of Vedic Mathematics-based instruction on Grade 7 students' performance and attitudes in solving problems involving polynomial operations. The topic was chosen due to its foundational role in transitioning from arithmetic to algebra. Poor mastery of polynomial operations often hinders higher-level mathematical understanding; hence, reinforcing this competency is crucial.

Vedic Mathematics is expected to provide benefits through its sutra-based techniques, offering simplified approaches to polynomial operations that may reduce cognitive load and lessen anxiety among learners. The intervention will involve three groups: two experimental groups exposed to Vedic Mathematics through Direct Instruction (DI) and Discovery Learning (DL), and one control group taught using traditional classroom methods. The same teacher will facilitate all groups to ensure consistency of instruction.

Respondents will be identified based on their performance in the Quarter 2 mathematics examination, specifically those scoring below 75%. The intervention itself will be conducted during

the third quarter of the 2025-2026 school year at Nasipit National Vocational School, Agusan del Norte, Philippines. This sequencing ensures that participant selection is grounded in prior performance while the instructional experiment is implemented in the subsequent quarter. Purposive sampling will be employed to identify a selected group of Grade 7 students who will participate in the study, rather than involving the entire Grade 7 population. The instructional materials will consist of lesson plans specifically developed by the researcher, and students' attitudes toward mathematics will be measured using a validated Likert-scale instrument. Data will be collected through pre-test, post-tests, and an attitude survey.

The study has several limitations. It will be conducted in a single secondary school with a relatively small sample (n = 60), limiting the generalizability of findings. Purposive sampling further restricts external validity, as participants are not randomly selected. Dependence on one teacher may introduce bias related to teaching style and classroom management. The focus on polynomial operations excludes other mathematics domains such as geometry, statistics, and trigonometry, which limits the scope of application. The intervention period of one quarter will not capture long-term effects on performance or attitudes. External factors such as prior knowledge, motivation, parental support, and home study environment may also influence outcomes beyond the instructional strategies applied.

Despite these limitations, the study is expected to make a meaningful contribution by systematically comparing instructional approaches in one area of mathematics learning. Findings will provide context-specific insights and evidence-based suggestions for teachers and policymakers seeking to improve mathematics instruction in Grade 7. The results may also serve as a foundation for future large-scale or longitudinal studies.

1.8 Definition of Terms

The terms below are used to clarify and unify the meanings of the variables and main ideas in this study. These definitions align with how each term is used when studying how Vedic Mathematics, taught through Direct Instruction and Discovery Learning, affects the math performance and attitudes of Grade 7 students.

The Attitude towards Mathematics. It refers to how students feel about mathematics—their confidence, motivation, anxiety, and perception of its usefulness. This will be measured using a 4-point Likert scale to capture respondents' feelings, motivation, and actions toward learning mathematics.

Mathematics Performance. It refers to how well respondents apply mathematical concepts and procedures correctly and efficiently. This will be measured using pre-tests and post-tests on problem-solving tasks to demonstrate how students' thinking skills interact with teaching methods.

Vedic Mathematics (VM). It refers to a set of mental computations established by Bharati Krishna Tirthaji, consisting of sixteen mathematical sutras designed to simplify arithmetic and algebra. In this study, VM will be incorporated into classroom instruction to enhance computational fluency, improve problem-solving accuracy, and boost learner confidence.

Direct Instruction (DI). It refers to a formal, teacher-structured instructional model involving clear explanations, model demonstration, systematic practice, and immediate feedback. In this research, DI will be used to teach Vedic Mathematics techniques to encourage procedural mastery and calculation accuracy.

Discovery Learning (DL). It refers to a learner-centered teaching model that emphasizes facilitated exploration, logical thinking, and inductive questioning. In this research, DL will engage respondents in exploring mathematical patterns and relations in Vedic Mathematics through self-performed problem-solving tasks that support conceptual cognition.

Experimental Group. It refers to the two target groups of Grade 7 participants who will be taught using Vedic Mathematics through Direct Instruction and Discovery Learning. Their performance and achievement will be compared using pre-test and post-test results.

Quasi-Experimental Design. It refers to the research design used to establish cause-and-effect relationships of instructional strategies without random allocation of participants. In this study, intact Grade 7 classrooms will be pre-tested and post-tested to determine the effects of the interventions on students' performance and attitudes toward mathematics.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter discusses the methods used in gathering the data for this research; the research design used, the study participants, the study's setting, the research instruments and how these were validated, the procedures undertaken, and the data analyses.

2.1 Research Design

This study will employ a quasi-experimental pretest-posttest non-equivalent groups design to examine the effects of integrating Vedic Mathematics on Grade 7 respondents' performance and attitudes in mathematics. This design is appropriate because it involves intact classes and does not require random assignment. Instead, purposive sampling will be used to select respondents identified as slow learners based on their performance in the previous quarter's mathematics examination, specifically those who scored below 75 percent.

The design enables comparison between groups by administering a pre-test to establish baseline performance, implementing the instructional intervention, and then conducting a post-test to measure achievement gains. An attitude survey will also be administered to assess changes in motivation, confidence, and attitudes toward mathematics. This structure ensures systematic measurement of both cognitive and affective outcomes while maintaining ecological validity in a real classroom setting.

2.2 Research Materials and Method

The instructional materials for this study will consist of two lesson plans developed by the researcher, both incorporating Vedic Mathematics techniques. One lesson plan will be based on the Direct Instruction model, which provides explicit teaching and guided practice to help respondents master mathematical procedures.

The other lesson plan will follow the Discovery Learning model, encouraging respondents to explore mathematical concepts through problem-solving and pattern recognition using Vedic techniques. These instructional approaches are designed to address the needs of Grade 7 respondents who scored below 75% in the Quarter 2 mathematics examination by providing structured and engaging learning experiences. Both lesson plans will be aligned with the MATATAG Curriculum and will focus specifically on operations on polynomials, a key competency in Grade 7 mathematics.

To ensure fairness and consistency, the researcher will personally facilitate instruction for both experimental groups. Each group will receive equal classroom time, identical instructional materials, and comparable learning conditions. This approach minimizes external variables and ensures that any observed differences in outcomes can be attributed to the instructional method itself.

Expert validation of the lesson plans and assessment tools will be conducted to ensure content accuracy, readability, and alignment with learning goals. Four experts will serve as validators: a mathematics specialist, a psychometrician, the research adviser, and the Guidance Coordinator of the school clinic. The research adviser's role will be limited to confirming that the content corresponds with the study's objectives, without directly influencing the validation process, in order to avoid potential bias. Feedback from the validators will be carefully considered and incorporated into the final versions of the instruments and lesson plans.

Data Gathering Procedure

The study will follow a transparent and ethical data collection process to ensure accuracy, reliability, and trustworthiness of the findings. The procedure will consist of four main phases: preparation and coordination, instrument validation, experimental implementation, and data consolidation. Each phase will adhere to the approved research plan and ethical standards.

Preparation and Gathering

Prior to data collection, the researcher will first obtain formal permission from the Principal of Nasipit National Vocational School (NNVS) in Nasipit, Agusan del Norte, as well as approval from the School Research Committee of Saint Michael College of Caraga (SMCC). Second, the Grade 7 mathematics teachers and advisers will be informed and consulted regarding the study, and consent forms will be distributed and collected in compliance with Republic Act No. 10173, also known as the Data Privacy Act of 2012. Third, respondents who scored below 75% in the Quarter 2 mathematics examination will be purposively selected. Fourth, the pre-test will be administered to establish baseline performance. Fifth, the intervention will be implemented through Direct Instruction or Discovery Learning. Sixth, the post-test will be conducted after completion of the targeted competencies. Finally, the Mathematics Attitude Inventory will be administered to measure affective outcomes. This structured sequence ensures systematic and transparent measurement of both cognitive and attitudinal changes.

Instrument Development and Validation

Two instruments will be used in the study: the (1) Mathematics Performance Test (MPT) and the (2) Mathematics Attitude Inventory (MAI). The MPT is a researcher-made instrument designed to assess respondents' mastery of Grade 7 mathematics competencies, particularly in algebra and polynomial operations. The MAI is an adapted and modified instrument intended to evaluate respondents' confidence, motivation, and anxiety toward mathematics. Both instruments will be tailored to the developmental level of Grade 7 learners and aligned with the MATATAG Curriculum.

Validation of the instruments will be conducted by four experts: a mathematics specialist, a psychometrician, the research adviser, and the Guidance Coordinator of the school clinic. The

experts will use a four-point Likert scale to evaluate each item for clarity, relevance, and alignment with the study's objectives. Items with a content validity index below 0.80 will be revised or removed to ensure the instruments meet acceptable standards of content validity. The research adviser's role will be limited to confirming the alignment of the instruments with the study's goals, without influencing the validation outcomes.

To establish reliability, a pilot test will be conducted at Saint Michael College of Caraga, involving Grade 7 respondents who will not be part of the main study. Reliability will be assessed using Cronbach's alpha, with coefficients expected to approximate 0.87 for the MPT and 0.90 for the MAI, indicating strong internal consistency and suitability for use in the main study.

Experimental Implementation

The quasi-experimental phase of the study will be conducted over one month during the second quarter of the 2025–2026 school year. Two intact Grade 7 classes, comprising a total of approximately 60 students identified based on their previous quarter's mathematics performance, are proposed to participate in the intervention. One group, designated as Experimental Group A, will receive instruction in Vedic Mathematics through Direct Instruction, focusing on structured guidance and practice. The second group, Experimental Group B, will be taught Vedic Mathematics through Discovery Learning, emphasizing inquiry and conceptual exploration. This arrangement will allow for a systematic comparison of two distinct instructional approaches within the same mathematical competency.

At the conclusion of the intervention, both groups will take the post-test simultaneously. Answer sheets will be collected promptly and stored securely to maintain anonymity and data integrity throughout the process.

Data Consolidation and Verification

Data analysis will be conducted using SPSS version 26. All responses will be coded, and frequency checks will be performed to identify missing or inconsistent data. The researcher will oversee the verification of data integrity, with the assistance of a statistician to ensure accuracy in the analysis. Anonymization techniques will be applied to protect the confidentiality and identities of all participants, in accordance with ethical research standards.

2.3 Research Locale (Research Environment)

The research will be conducted at Nasipit National Vocational School (NNVS), located in Barangay 7, Nasipit, Agusan del Norte. Established in 1967 as a community-based trade school, NNVS has evolved into a recognized public secondary institution offering academic and technical-vocational programs aligned with TESDA standards. The school actively participates in regional educational initiatives such as the Regional Assessment of Learning and the Senior High School Summit, reflecting its commitment to technical excellence and holistic learning.

NNVS implements the MATATAG Curriculum prescribed by the Department of Education, emphasizing basic literacy, numeracy, and lifelong learning. Its blend of academic and technical programs supports the national goal of competence-based education. With a multicultural student population and faculty, NNVS provides a diverse learning environment. Classrooms reflect typical Philippine public-school conditions, often marked by resource constraints and challenges in sustaining student motivation. This context presents an ideal setting for testing innovative instructional approaches such as Vedic Mathematics-Based Instruction. The school's willingness to support instructional enhancements ensures administrative and faculty cooperation throughout the study.

2.4 Research Respondents, Population, and Sample

This study will involve Grade 7 students enrolled at Nasipit National Vocational School during the School Year 2025–2026. A quasi-experimental design will be employed, utilizing intact classes to preserve natural learning conditions while comparing instructional methods.

The respondents will be selected through purposive homogeneous sampling, targeting students identified as slow learners based on their performance in the Quarter 2 mathematics examination, specifically those who scored below 75 percent. This sampling technique ensures that the selected participants share similar academic characteristics, allowing for equitable treatment and meaningful instructional comparisons.

Two intact classes of students are proposed for inclusion in the intervention, with each group consisting of approximately 25 to 30 students (total sample size: 50–60). This sample size is considered manageable and sufficient for classroom-based quasi-experimental research. It supports hypothesis testing at the 0.05 level of significance, although the study acknowledges that statistical power may be limited, and results should be interpreted with caution regarding generalizability.

The first experimental group will receive instruction in Vedic Mathematics through Direct Instruction, which emphasizes guided teaching and structured practice. The second experimental group will study Vedic Mathematics through Discovery Learning, which encourages inquiry, exploration, and conceptual understanding. The intervention will be implemented over the course of one month and will focus on a single mathematics competency, specifically operations on polynomials. To ensure fairness and reliability, both groups will be provided with equal time allocations, standardized instructional materials, and consistent classroom conditions.

2.5 Research Data and Analysis Plan

The order of data collection will be strictly observed: selection of participants based on Quarter 2 exam results, administration of the pre-test, implementation of the intervention, conduct of the post-test after competency completion, and administration of the attitude questionnaire. This sequence will ensure consistency in data gathering and validity in comparing outcomes across groups.

Data from the pre-test, post-test, and attitude survey will be analyzed descriptively and inferentially to assess the impact of Vedic Mathematics instruction on students' performance and attitudes. Performance and attitude levels, both before and after the intervention, will be summarized using descriptive statistics, including mean, standard deviation, and percentage. To determine significant differences in performance and attitude between groups exposed to Direct Instruction (DI) and Discovery Learning (DL), a One-Way Analysis of Variance (ANOVA) will be conducted on post-test mean scores.

Additionally, paired sample t-tests will be used to compare pre-test and post-test results within each group, assessing improvements after the intervention. The Pearson product-moment correlation coefficient will be computed to examine the strength and direction of the relationship between mathematics performance and attitudes toward mathematics.

All statistical analyses will employ the 0.05 level of significance to guide decisions on the null hypothesis. Given the proposed sample size of 25 to 30 students per group, the study acknowledges the risks of Type I error (rejecting a true null hypothesis) and Type II error (failing to detect a real effect). Although the sample size is considered sufficient for classroom-based quasi-experimental research, statistical power may be limited, particularly for detecting small

effect sizes. Therefore, findings will be interpreted with caution and viewed within the scope of classroom-level experimentation.

2.6 Ethical Standards

The ethical standards in this study will be guided by the policies of the Department of Education (DepEd) and aligned with the institutional requirements of Saint Michael College of Caraga (SMCC). All procedures will be approached responsibly and transparently, with maximum respect for the rights and well-being of the respondents. Ethical compliance will be ensured by adhering to the research guidelines established in the Division of Agusan del Norte Research Manual.

Institutional approval will be secured through the School Research Committee of SMCC, while formal permission to conduct the study will be obtained from the Principal of Nasipit National Vocational School (NNVS), the designated research locale. The study will uphold principles of confidentiality, voluntary participation, and informed consent, ensuring that respondents and their guardians are fully aware of their rights and protections throughout the research process.

A. Protection of Intellectual Property Rights (IPR)

The study will emphasize the significance of protecting intellectual property rights by ensuring that all research outputs, including tools and analytical frameworks, comply with Philippine copyright law, specifically Republic Act No. 8293. This will entail that any original materials produced, such as surveys or interview questions, are meticulously designed to prevent any infringement. All third-party sources and referenced literature will receive proper attribution, thereby protecting the rights of all contributors. By following these guidelines, the research will uphold the intellectual property of others and reinforce the credibility and integrity of the study.

B. Informed Consent

Respondents, who will be students from NNVS, will be provided with comprehensive information regarding the research objectives, methodologies, anticipated outcomes, and potential benefits. The researcher will clearly outline what participation entails, including the nature of the questions and the estimated time commitment. To ensure that respondents fully understand their role and how their responses will be used, written informed consent will be obtained from all human participants and their guardians before participation.

The use of animals or other living organisms does not apply to this study, as it involves only human respondents. Ethical approval will be secured from the institution before implementation, in accordance with research protocols.

C. Data Privacy and Confidentiality

The utmost confidentiality will be observed in all information gathered. A numeric code will be assigned to respondents to maintain anonymity, and no identifying information will be released. Data will be stored in password-protected files, accessible only to the researcher and thesis adviser. Upon completion of the study, all raw data will be destroyed in accordance with institutional data-retention policies. Results will be presented only in summary form to avoid accidental disclosure.

D. Voluntary Participation and Freedom to Withdraw

Participation in this study will be entirely voluntary. Respondents and their guardians will be provided with informed consent documents and will be informed that they may withdraw from the study at any time without penalty or disadvantage.

No coercion will be applied, and opting out will not affect their academic standing or relationship with the school. The use of animals or other living organisms is not applicable to this study, as it involves only human respondents.

E. Minimization of Harm and Risk Management

Respondents' risk will be minimal, as all activities will be conducted under normal classroom conditions as part of the regular educational process. The researcher will ensure that no respondent experiences physical, psychological, or emotional harm. Teaching sessions will be overseen by the subject teacher in accordance with the Child Protection Policy of DepEd. Should any respondent experience discomfort, they may report it to the researcher, the cooperating teacher, the school guidance counselor, or the school clinic. Appropriate support will be provided, and if necessary, withdrawal from the study will be facilitated without academic consequence. To minimize harm, available school-based support services such as counseling, health consultation, or academic assistance will be applied as needed, in coordination with institutional protocols. The use of animals or other living organisms does not apply to this study, as it involves only human respondents.

F. Beneficence and Contribution to Knowledge

The proposed study should be beneficial both in the short and long term. In the short term, students will be exposed to Vedic mathematics, which will give them confidence and fluency in mathematics. As respondents, they will directly benefit from engaging with culturally enriched mathematical strategies that may enhance their numeracy skills, problem-solving abilities, and classroom participation. The findings can be added to the growing body of literature on the pedagogy of mathematics, particularly on how to incorporate the vast wealth of cultural mathematical practices into modern pedagogical systems.

The findings can also assist teachers, administrators, and policymakers in developing and informing the formulation of curriculum and instruction to better meet the numeracy objectives of the MATATAG Curriculum. By evaluating the combined use of Vedic Mathematics, DI, and DL, the study contributes to evidence-based instructional reform aligned with the MATATAG Curriculum. The research findings will be made available upon the request of the respondents. The use of animals or other living organisms is not applicable to this study, as it involves only human respondents.

G. Justice and Fair Participant Selection

Participants will be selected using purposive homogeneous sampling to ensure fairness and relevance to the study's objectives. Selection will be based solely on the criterion of scoring below 75% in the Quarter 2 mathematics examination. Within this group, no student will be favored or excluded based on gender, socioeconomic status, or other personal characteristics, thereby ensuring equitable treatment of all eligible participants. The use of animals or other living organisms does not apply to this study, as it involves only human participants.

H. Data Integrity and Accuracy

The researchers will ensure that all data recorded is accurate, reliable, and faithfully represented. Raw scores will be carefully encoded and verified prior to statistical analysis using the Statistical Package for the Social Sciences (SPSS). No data manipulation or fabrication will be undertaken. Interpretation will remain objective, guided strictly by the research objectives and aligned with the intended learning outcomes.

To mitigate potential bias, errors, and limitations, the researcher will implement double-checking procedures during data encoding, apply consistent scoring rubrics, and transparently report any anomalies encountered during analysis. If discrepancies or unclear responses arise, the researcher will respectfully reach out to the concerned respondents for clarification, ensuring that their privacy and consent are upheld throughout the process.

I. Transparency and Honesty in Reporting

There will be no distortion of data or hiding. The researcher will provide detailed descriptions of the methods, procedures, and limitations. The assistance, regardless of the form, such as expert validation or software computation, will not be neglected. Such transparency will be replicable and credible.

All materials, sources, and references used in the study will be properly cited to uphold academic integrity. If artificial intelligence tools are used to assist in writing, analysis, or editing, their role will be clearly acknowledged in the manuscript. To mitigate potential conflicts of interest, the researcher will disclose any affiliations, funding sources, or external influences that may affect the interpretation or reporting of results.

J. Use of Patented or Copyrighted Materials

All instructional and visual materials used in the study will be properly cited in accordance with the APA 7th Edition. No patented or copyrighted materials requiring a license or commercial use will be employed. The methods of Vedic Mathematics are in the public domain; however, any contributions or interpretations by contemporary scholars will be duly acknowledged and credited.

K. Ethical Considerations for Human Trials

The research will involve student participants and will be conducted in accordance with DepEd Order No. 40, s. 2012 (Child Protection Policy). The researcher will ensure that participation

does not lead to coercion, discrimination, or inequitable treatment. All instructional procedures and research activities will be reviewed and approved by the school administration prior to implementation, ensuring ethical compliance and institutional alignment.

Respondents will be fully informed of the potential risks and benefits of participating in the study, and their voluntary consent will be obtained. Humane treatment standards will be upheld throughout the research process, ensuring that all participants are treated with dignity, fairness, and respect. Should any discomfort or distress arise, school-based support services such as counseling, health consultation, or academic assistance will be made available in coordination with institutional protocols, reinforcing the study's commitment to minimizing harm.

L. Responsible Use of AI and Other Related Technologies

The researcher ensured the ethical use of electronic tools throughout the study. Grammarly was employed to refine grammar, spelling, and style, while ChatGPT was used strictly as a supportive tool for improving clarity, formatting, and organization of text. Neither application was utilized to generate original research content, data, or ideas. For statistical analysis, SPSS was applied, while Microsoft Excel and Word were used for data processing, tabulation, and documentation. All tools were used responsibly, with the researcher maintaining full authorship and accountability for the study's content.

M. Ethical Clearance and Institutional Approval

Before data collection, the researcher will secure ethical clearance from Saint Michael College of Caraga (SMCC) to ensure compliance with academic and moral standards. Since the study will be conducted at Nasipit National Vocational School (NNVS), administrative approval from the Office of the Principal of NNVS will also be obtained. This two-step process guarantees

that the research adheres to the ethical requirements of the researcher's home institution while respecting the authority and policies of the cooperating school.

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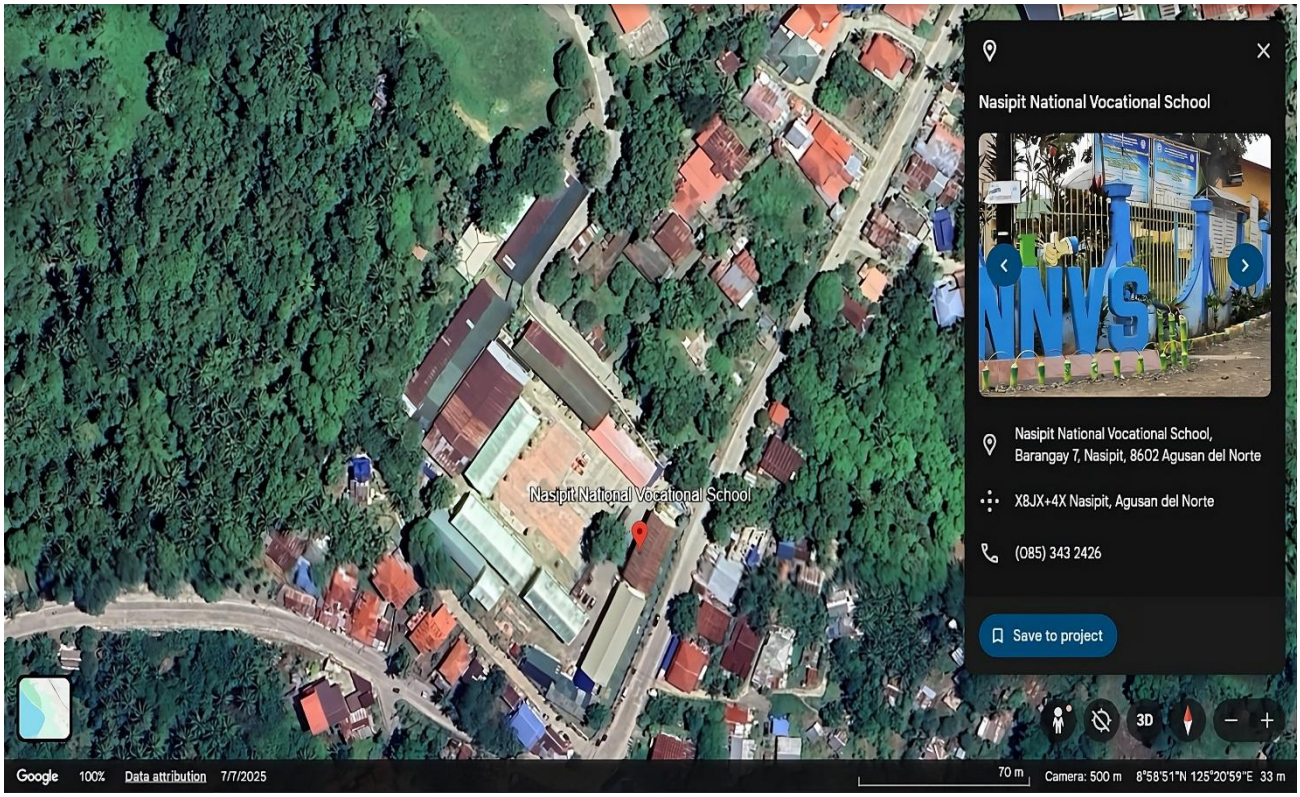
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APPENDICES

A - Letter of Approval with Certificate of Appearance and Photo

B - Instrument with Informed Consent

C - Map of the Research Locale *(Include Coordinates)*



D - Narrative & Photo Documentation
E- Ethics Committee Approval

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